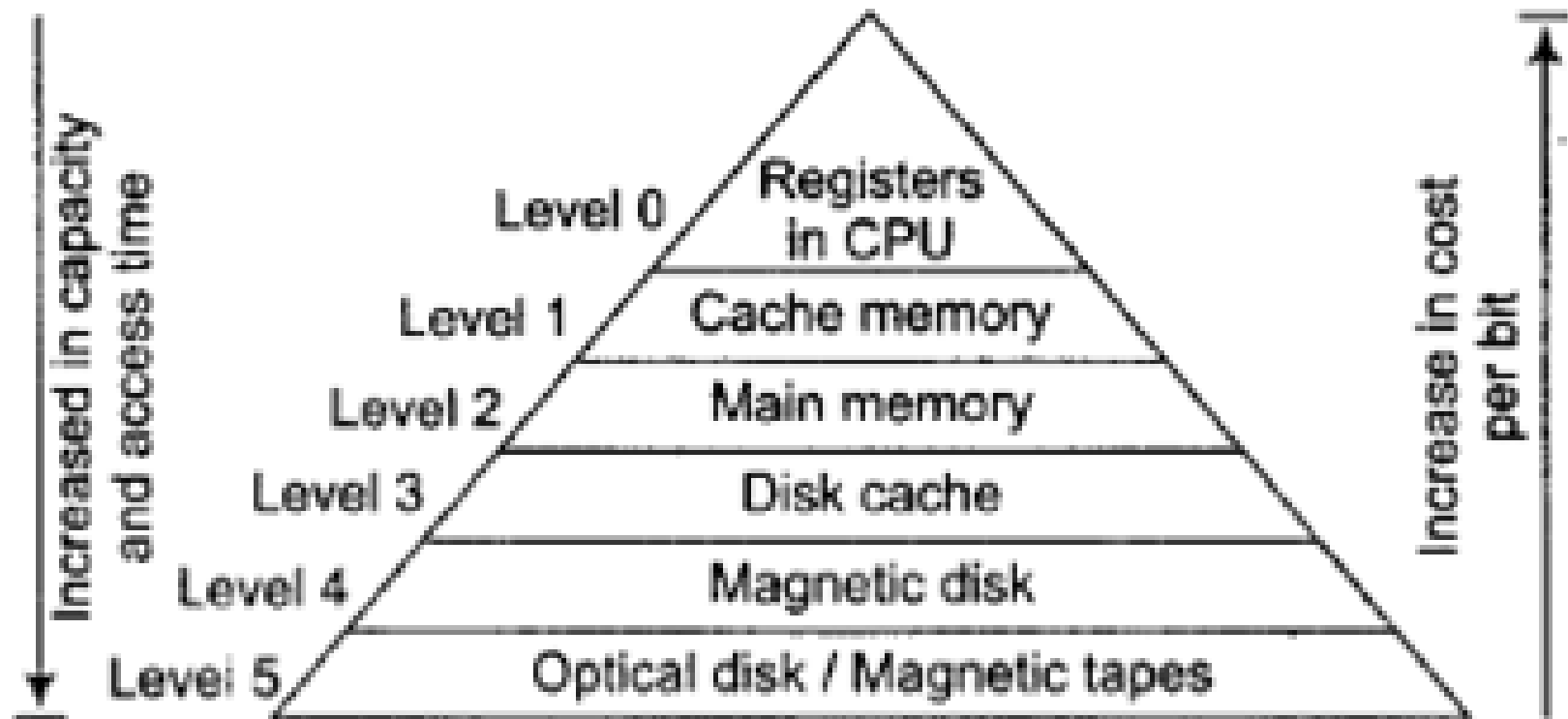


Memory Hierarchy



Differences between compiler and interpreter

What is a Compiler in computer programming?

A compiler is a computer program that reads statements in one programming language and translates them into another language that a computer can process. Most computer programmers write lines of code on their own using programming languages like C or Pascal, which can create a collection of independent source statements. The compiler then evaluates each of these statements on its own and passes them on together to the next code stage by making sure that each line of code is referenced accurately after being compiled. After it undergoes compiling, the original code becomes object code, sometimes also called machine code.

Most compilers use programming languages like Java, C++ and C.

Differences between compiler and interpreter

What is a Interpreter in computer programming?

An interpreter is a program that functions similarly to a compiler by translating lines of code into machine or object code. However, interpreters typically translate lines of code or instructions one at a time, as opposed to all at once, as a compiler does. Interpreters also create new lines of source code each time they operate, instead of producing one single batch of source code at the end of interpreting. This can mean that interpreted code sometimes runs slower than compiled code, as it creates new source code with every use.

Some of the most popular programming languages that interpreters can use include Python, Ruby and JavaScript.

Differences between compiler and interpreter

Compiler	Interpreter
• A compiler takes the entire program in one go.	• An interpreter takes a single line of code at a time.
• The compiler generates an intermediate machine code.	• The interpreter never produces any intermediate machine code.
• The compiler is best suited for the production environment.	• An interpreter is best suited for a software development environment.
• The compiler is used by programming languages such as C, C ++, C #, Scala, Java, etc.	• An interpreter is used by programming languages such as Python, PHP, Perl, Ruby, etc.

Input Output Module's Key Tasks

- 1.Processor communication** – This involves a number of tasks, primarily the transference of data between the processor and an I/O module, accepting and decoding commands sent by the processor, reporting of current status, and an ability for the I/O module to recognise its own unique address.
- 2.Device communication** – It needs to be able to perform standard device communications, such as reporting of status.
- 3.Control and timing** – An I/O module needs to be capable of managing data flow between a computer's internal resources and any connected external devices.
- 4.Data buffering** – A crucial function that manages the speed discrepancy that exists between the speed of transfer of data between the processor and memory and peripheral devices.
- 5.Error detection** – Detecting errors, whether mechanical (such as a printer experiencing a paper jam) or data based, and reporting them to the processor is another vital function of an I/O module.

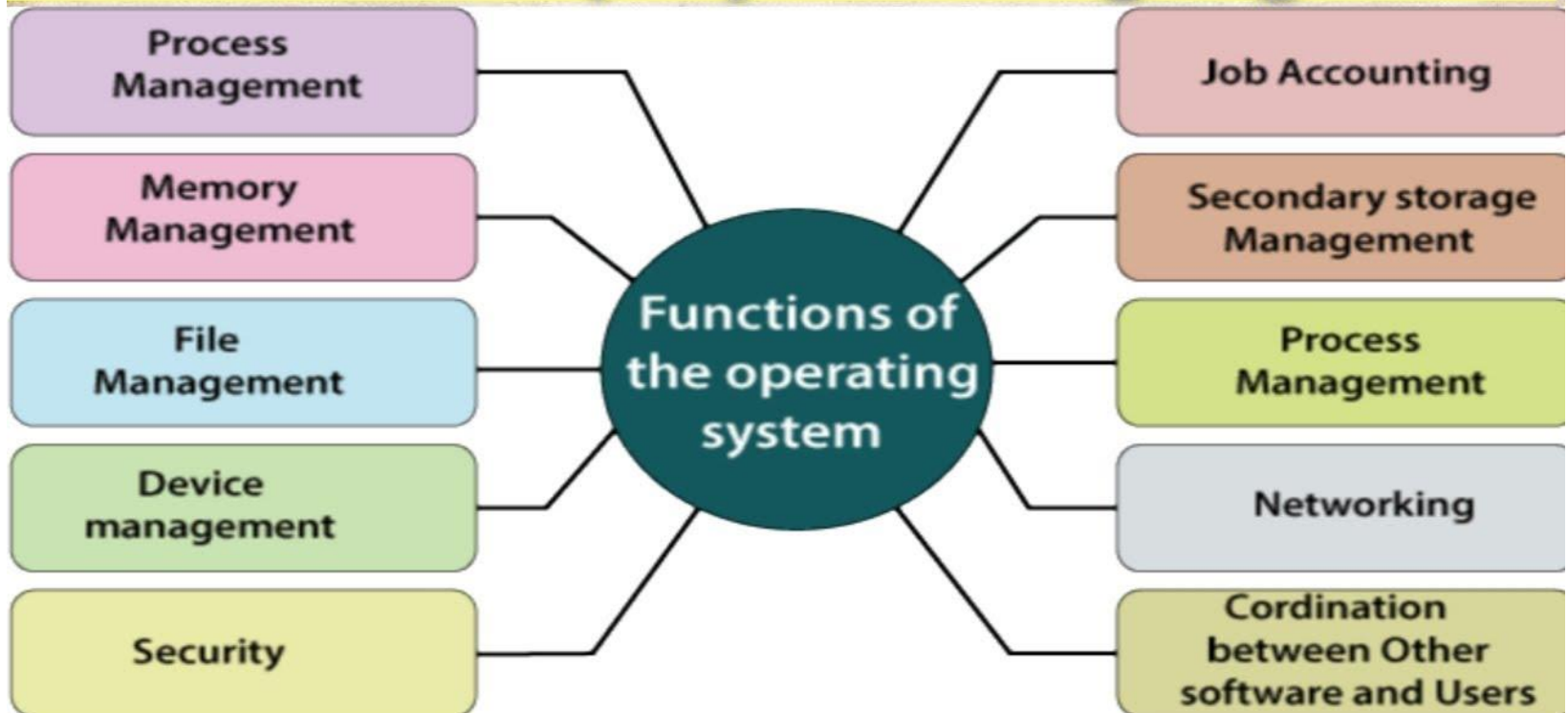
Functions of an Operating System

What is an Operating system?

- An operating system is the primary software that manages all the hardware and other software on a computer. The operating system, also known as an “OS,”
- Operating Systems provides for the users to utilize the functions of a computer by managing the hardware and the software

Functions of an Operating System

Functions of operating System



Copyright

What is copyright?

- Copyright refers to the legal right of the owner of intellectual property. In simpler terms, copyright is the right to copy. This means that the original creators of products and anyone they give authorization to are the only ones with the exclusive right to reproduce the work.
- Copyright Law gives creators of original material the exclusive right to further use and duplicate that material for a given amount of time, at which point the copyrighted item becomes public domain.

Copyright

WRONGS AND RIGHTS

What you can and can't do under the amended Copyright Act

Can do

- Share copyrighted photos through social media for personal use, though credit to copyright holder is required
- Use copyrighted photos for education purposes such as reports or homework, though credit is required



Can't do

- Delete copyright credits
- Use copyrighted photos for internal meetings or presentations in an organisation
- Use copyrighted photos for commercial purposes in social media, business websites or other channels such as billboards, books, T-shirt screens, advertising without a credit
- Edit a copyrighted photo for commercial purposes without permission

What is software piracy

- Software piracy is the unauthorized downloading, copying, use, or distribution of software. Downloading and using software without paying for it is a common tactic of pirated software users.
- However, software piracy also includes distributing software on multiple machines when a license was only purchased for one, as well as copying software and redistributing it.

Role of IT in Society

What is Information Technology?

- Information technology is the technology used to store, manipulate, distribute or create information. All these can be summed up easily – It's having knowledge, and knowledge comes from having information. Gaining knowledge through information is the role of "**information technology**" IT in today's informed world.

Roles of IT in Society

Positive impact

- Communication
- Data Management
- Improved Education and learning Process
- Mechanized Agriculture
- Easy to access Information
- Globalization
- Creating new jobs

Negative impact

- Increase of health and mental concerns
- Increase of cybercrime
- Game addiction
- Identity theft